

Scalability & Future Plan

Maximum Marks: 1 Mark

Date	Team ID	Project Name
15 March 2026	Batch 2023-27 ACOE	Rising Waters – ML-Based Flood Prediction System

Team Members

S.No	Team Member Name	Role
1	Venkatesh Balireddy	Team Lead
2	Archana Dhanani	Member
3	Shivatmika Gandikota	Member
4	Gode Siva Ramakrishna Durgaprasad	Member
5	Bavisetty Gopi Krishna	Member

Current Scalability Architecture

Rising Waters is architected to scale horizontally using Docker containers on IBM Cloud Foundry. The stateless Flask application can run as multiple instances with a shared prediction log.

Aspect	Current Implementation	Scale-Up Path
Hosting	IBM Cloud Foundry (single instance)	Scale to multiple CF instances
Container	Docker (Dockerfile provided)	Kubernetes orchestration via IBM CKS
Storage	CSV flat file (predictions_log.csv)	PostgreSQL or IBM Db2 cloud database
Model	XGBoost (batch retrain)	Online learning with new incoming data
Frontend	Flask-rendered Jinja2 templates	React/Vue SPA with Flask REST API backend

Future Enhancement Roadmap

S.No	Enhancement	Description	Priority	Assigned To
1	Real-time API Integration	Connect to IMD / OpenWeatherMap API for live meteorological data	High	Venkatesh Balireddy
2	SMS / Email Alerts	Auto-notify registered authorities when flood risk is detected	High	Archana Dhanani

S.No	Enhancement	Description	Priority	Assigned To
3	Multi-district Batch Pred.	Batch predict flood risk for multiple districts simultaneously	Medium	Shivatmika Gandikota
4	LSTM Time-Series Model	Replace XGBoost with LSTM for sequential rainfall pattern analysis	Medium	Gode S R Durgaprasad
5	Mobile Application	Native Android / iOS app using the Flask REST API backend	Medium	Bavisetty Gopi Krishna
6	Regional Language UI	Multi-language support for Telugu, Hindi for local disaster officers	Low	All Members
7	Satellite Data Integration	Add satellite imagery flood features via IBM Watson Visual Recognition	Low	All Members